

Shishir Adhikari

AI Engineer | RAG, LLM & Computer Vision Systems | FastAPI, PyTorch, LangChain

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SUMMARY

AI Engineer building production machine learning systems -- Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), computer vision, and NLP -- deployed as scalable REST APIs with FastAPI, Flask, and Django. Experience spans RAG pipelines, Nepali/Devanagari OCR, computer-vision face- and skin-analysis models (detection, segmentation, classification), and LLM-powered backend services using PyTorch, LangChain, and vector databases (FAISS, ChromaDB, pgvector).

EXPERIENCE

AI/ML Engineer Intern -- Nep Tech Pal

Nov 2025 - Apr 2026 | Pokhara, Nepal

- Developed and deployed production REST APIs (FastAPI, Flask, Django) serving AI-powered products to real users.
- Built computer-vision face- and skin-analysis pipelines: face-shape detection, acne severity (DINOv2), acne detection (YOLO/RT-DETR), wrinkle segmentation (U-Net++/EfficientNetV2), and age estimation (ViT).
- Engineered OCR systems for scanned documents and images, including Nepali-language text extraction.
- Built a real-time facial emotion-recognition system (ViT + OpenCV) detecting 8 expression classes through an interactive GUI.
- Developed an AI property-search system matching listings to user preferences (price, location, features).
- Improved backend reliability with structured logging, error handling, and async optimization.

PROJECTS

Face Shape Analyser

Flask | PyTorch | MediaPipe | OpenAI API | Google Gemini API

- Real-time computer-vision API classifying faces into 5 shapes at 85% accuracy; dual-LLM fallback (OpenAI GPT + Google Gemini) for 100% uptime, with Swagger docs and API-key auth.

Shikshya Chautari

FastAPI | LangChain | FAISS | AWS S3

- RAG exam-question forecasting system over 1,000+ pages: 0.827 BERTScore F1, 89.3% syllabus precision across 44 held-out papers, sub-1.5s responses, 2.8% hallucination rate.

Nepal Parichaya

Django | ChromaDB | Google Document AI | Streamlit

- Multilingual RAG assistant over Nepali source material; Document AI OCR (43 pages to 950+ chunks), hybrid ChromaDB + Devanagari keyword retrieval, Recall@5 of 0.80, 100% out-of-scope refusal.

Conversational-RAG API

FastAPI | Redis | LangChain | GPT

- Production conversational-AI backend with Redis-backed session memory (<50ms retrieval) and ~1.2s end-to-end responses for multi-turn, context-aware answers.

EDUCATION

BSc Computer Science & Information Technology (CSIT)

Tribhuvan University (IOST) | 2022 - 2026

CERTIFICATION

Complete Generative AI with LangChain and HuggingFace -- Udemy, 2024

Credential ID: UC-2b2a53d7-1b67-46c7-bc74-00f917e9b045

SKILLS

Programming: Python, JavaScript, SQL, HTML, CSS

Frameworks: FastAPI, Flask, Django, Streamlit, Scikit-learn

AI & ML: RAG, LLMs, NLP, OCR, Computer Vision (Detection, Segmentation, ViT), Fine-tuning, Prompt Engineering

Libraries: PyTorch, LangChain, LangGraph, HuggingFace, OpenCV, MediaPipe

Databases: PostgreSQL, MySQL, MongoDB, Redis, FAISS, ChromaDB, Pinecone, pgvector

Cloud & Tools: Docker, Git, REST APIs, AWS S3, Cloudflare